Regularization Techniques in Neural Network

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- ✓Methods of regularization
  - ✓Lasso and Ridge Regularization
  - ✓Data Augmentation
  - ✓Early stopping
  - ✓Drop out

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Regularization

Regularization is a set of methods for reducing **overfitting** in machine learning models.

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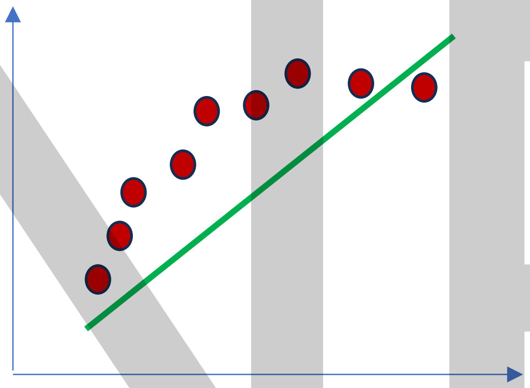
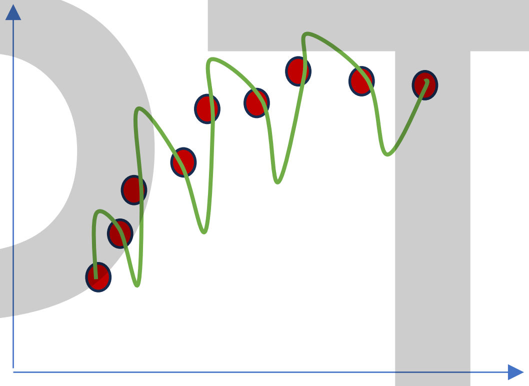
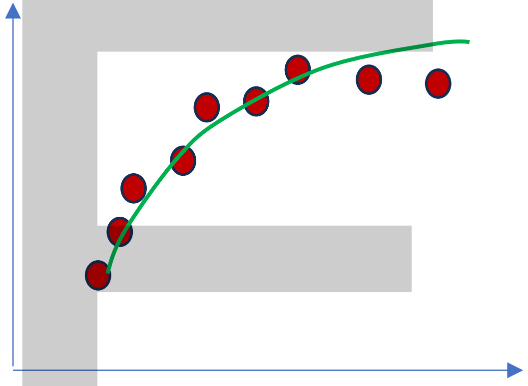


Underfitting



Overfitting



Correct fitting

Overfitting occurs when a machine learning model learns not only the true patterns in the training data but also the noise and random variations.

As a result, the model performs well on the training set but fails to generalize effectively to new, unseen data.

L1 Regularization- Lasso Regression

Least Absolute Shrinkage and Selection Operator

Least absolute error

Reducing some weights to 0

only keeping the key features within the model

Helps feature selection and reduce overfitting

Core Idea:

- ✓ Reduces overfitting by simplifying the model.
- ✓ Simplifies the model by shrinking some of the weights to zero.
- ✓ This is done by adding a penalty term to the loss function that is proportional to the sum of the absolute values of the model's coefficients (weights).

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L1 Regularization- Lasso Regression

$$L(w)_{\text{L1 regularized}} = L(w)_{\text{normal}} + \underbrace{\lambda \sum_i |w_i|}_{\text{Penalty term}} \longrightarrow \text{L1 norm}$$

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λ is a hyper parameter which decides the strength of regularization

L1 and L2 norm of a Vector

X=[-3,4,5,-2]

L1 norm of X

$|X|_1=|-3|+|4|+|5|+|-2|= 3+4+5+2=14$

L2 norm of X

$|X|_2=\sqrt{(-3)^2+(4)^2+(5)^2+(-2)^2}=7.34$

How L1 Regularization Helps in Feature Selection

$$L(w)_{\text{L1 regularized}} = L(w)_{\text{normal}} + \lambda \sum_i |w_i|$$

- The derivative of the penalty term in the loss function, is a constant
- Constant derivative means, L1 aggressively pulls the weight towards zero with the same strength until they become exactly zero.
- When any weight is zero, that feature is removed from the network

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L2 Regularization (Ridge Regression)

$$\mathrm{L(w)}_{\text{L2 regularized}} = \mathrm{L(w)}_{\text{normal}} + \underbrace{\lambda \sum_i w_i^2}_{\text{Penalty term}}$$

- Adds the squared value of each coefficient as a penalty term to the loss function.

Difference between L1 and L2 Regularization

Unlike L1 regularization, L2 regularization cannot reduce coefficient values exactly to zero. It can reduce the weight nearly to zero.

L2 regularization cannot perform feature selection.

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L2 Regularization Cannot Perform Feature Selection

In L2 regularization, the penalty term is

$$\lambda \sum_i w_i^2$$

The derivative of this penalty term w.r.t. weight is $2\lambda\omega_i$

- Force pulling the weight towards zero is proportional to the weights current value.
- When weight is close to zero, the force pulling it to zero also becomes very small
- This prevents it from reaching to zero.

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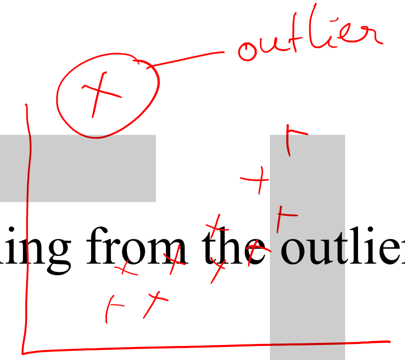
L2 Regularization is Not Robust to Outliers

$$L(w)_{\text{L2 regularized}} = L(w)_{\text{normal}} + \underbrace{\lambda \sum_i w_i^2}_{\text{Amplifies the error coming from the outlier points}}$$



Tries to adjust the model to accommodate these outliers

L1 Regularization

$$L(w)_{\text{L1 regularized}} = L(w)_{\text{normal}} + \underbrace{\lambda \sum_i |w_i|}_{}$$

Penalty is proportional to absolute values of weights and hence does not amplify errors coming from outliers

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Choice between L1 and L2 Regularization

Depends on the dataset and the final goal of analysis

If data is High dimensional with multiple features
Ex: Healthcare dataset

→ L1 regularization is preferred as it can do feature selection leading to a sparse model.

If dataset contains highly correlated features.(Multicollinearity)

Ex: Financial datasets

→ L2 regularization can be more suitable as it encourages small but non-zero coefficients.

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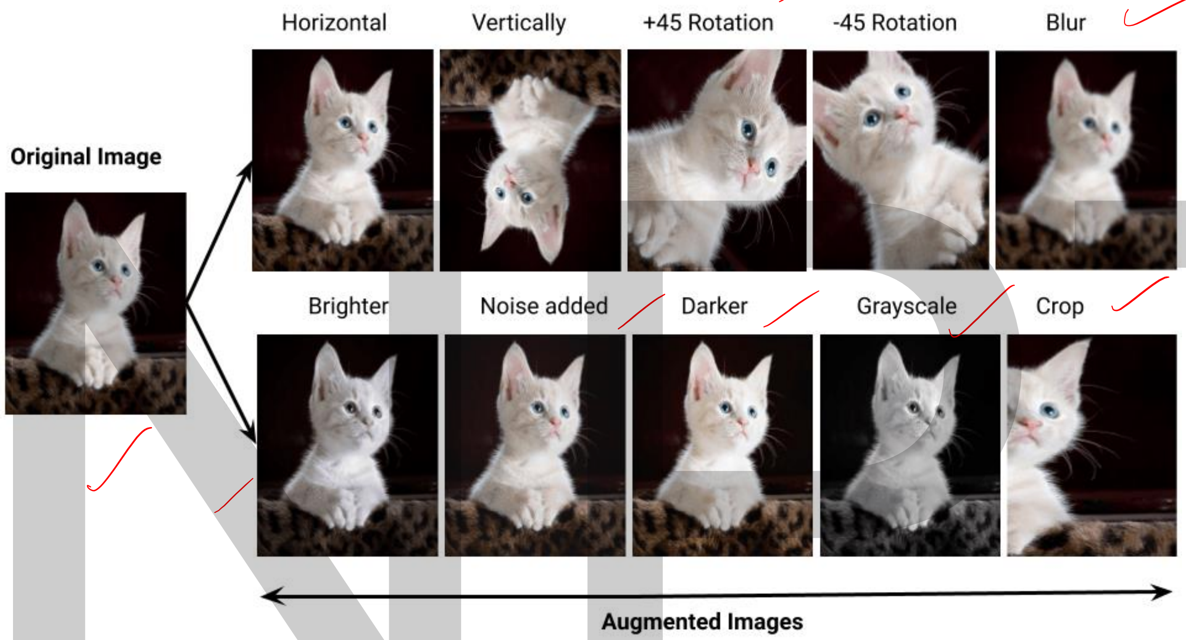
Data Augmentation



Creating new training samples by applying random transformations to your existing data
(label invariant transformation)

How it helps in Regularization?

- ➢ Neural network generalizes better as it gets exposed to more diverse set of training examples thus preventing overfitting.
- ➢ As deep neural networks require a large training dataset, data augmentation is also helpful when we have insufficient data to train a neural network.

Transformations

- Rotating
- Cropping
- Flipping
- Scaling
- Color space augmentation (brightness, saturation, contrast)

Image ref: https://ubiai.tools/what-are-the-advantages-anddisadvantages-of-data-augmentation-2023-update/

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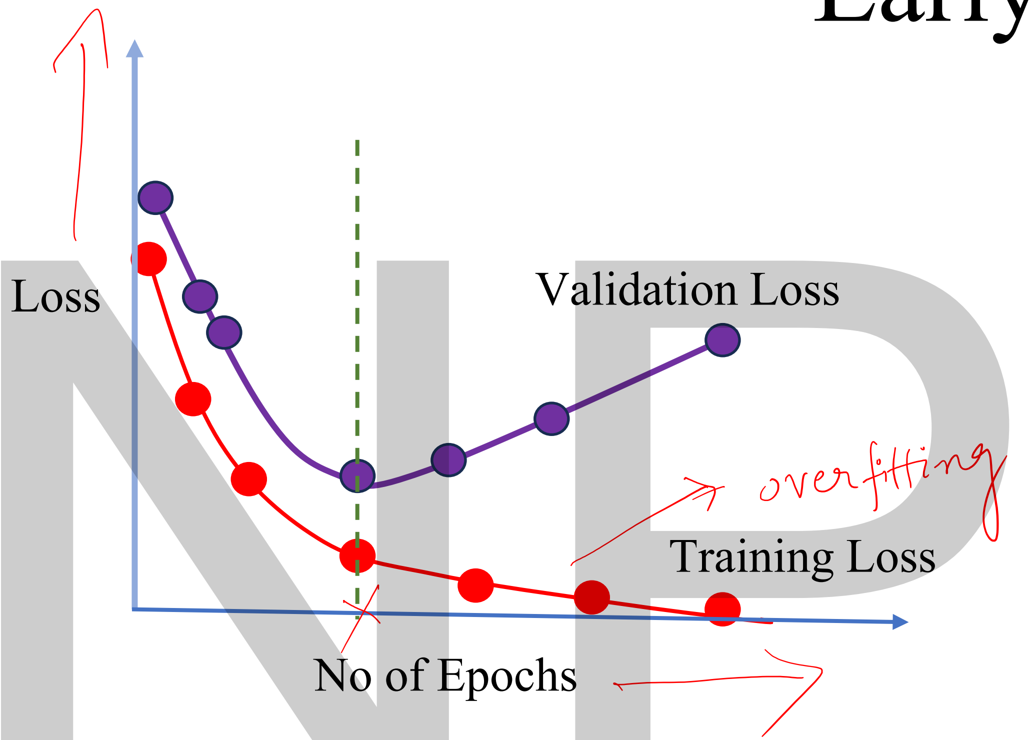
Early Stopping



- Early stopping monitors validation loss during training.
- If the validation loss fails to decrease for a predefined number of epochs (referred to as *patience*), training is halted.
- This ensures that the model is captured at the point where it achieves optimal performance on unseen data, preventing overfitting while reducing unnecessary computation.

Validation loss evaluates the model's performance on a portion of the dataset that the model has never seen during training.

- Run validation set evaluation periodically
- Choose stopping time based on validation loss

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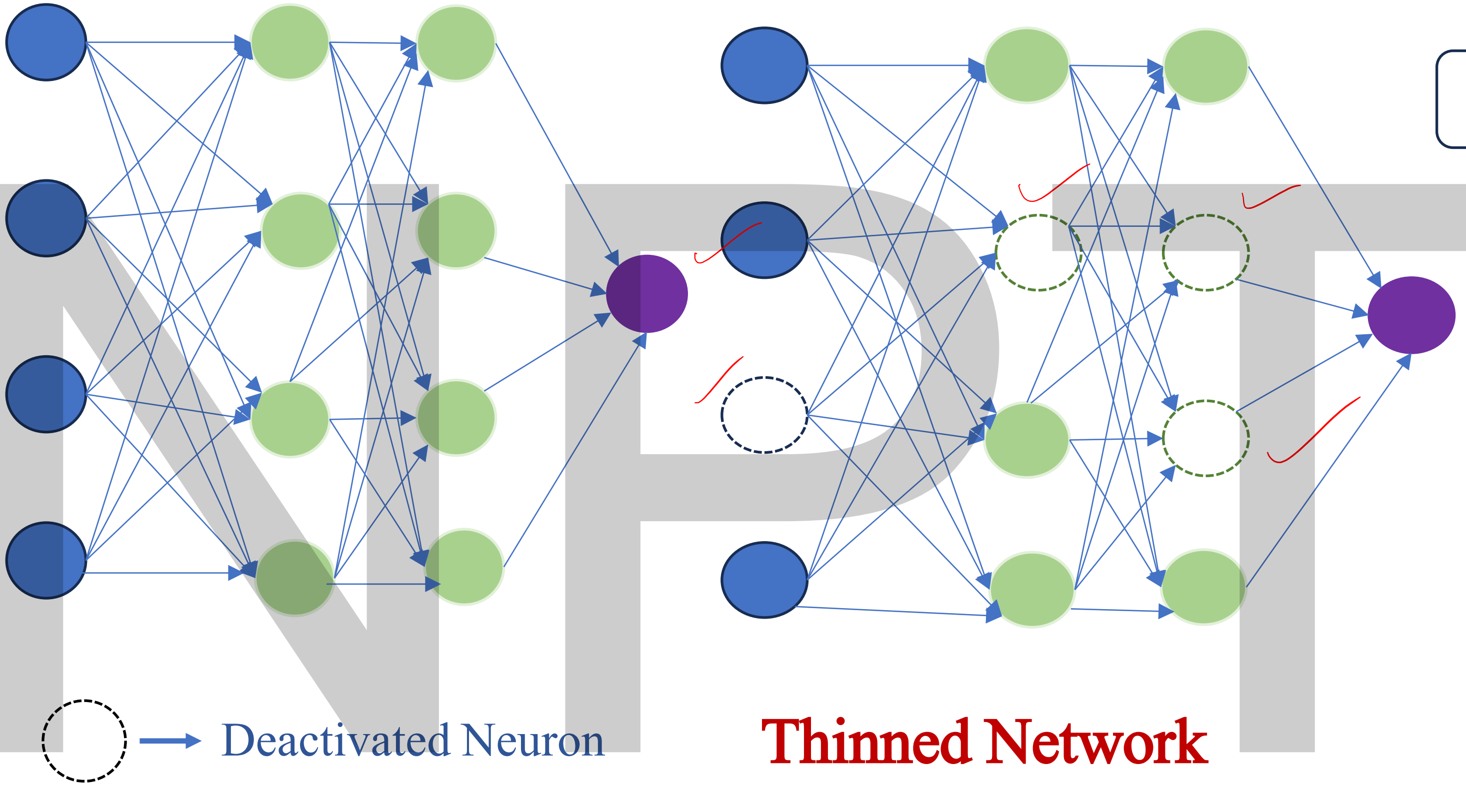
Dropouts



Training Phase

- During each iteration, randomly "drop" (set to zero) a fraction of neurons in the network with some probability
- Ensures that the network does not become overly reliant on particular neurons.

If there are n nodes in a network, then 2^n network configurations (thinned networks) are possible

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Process flow of Dropout Regularization

During Training

Define a dropout rate
or
dropout probability (*p*)

Create a dropout mask (D) for each layer with random values
(either 0 or 1)

Apply the mask to the layers activation (A) by element wise multiplication

This makes some neurons inactive in the network making it the first thinned network.

Scale the remaining activations by 1/(1-p)

Check which edges are present in this thinned network

Update the weights associated with only these edges

Iterate the process for another thinned network

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Dropout Regularization

During Testing

Deactivate the dropout mechanism → Do not apply any scaling

How Dropout is preventing overfitting?

- During training, neurons are removed randomly, thus preventing over tuning of any node to achieve the minimum cost function.
- An ensemble of small neural networks(thinned models with different structures) are trained during each iteration, which improves the model's capability to generalise to unseen data.

Limitations of Dropout regularization

- Increased training time

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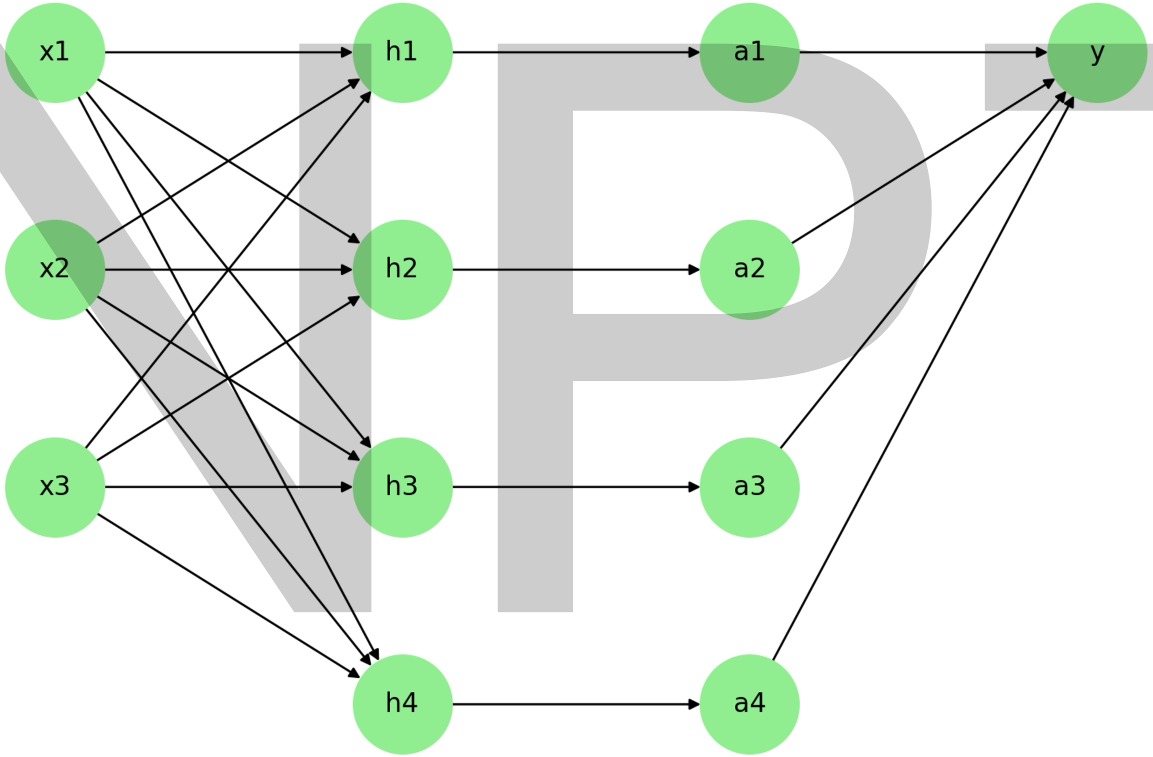
Example of Dropout Regularization



Let $X=[2, -1, 0.5]^{\top}$

Dropout probability (p)=0.2

Keep probability (1-p)=0.8

Let mask=[1,0,1]

After application of mask:

$X=[2,0,0.5]^{\top}$

With scaling:

$$\mathbf{X} = \frac{1}{0.8}([2,0,0.5]^{\top})=[2.5,0,0.625]^{\top}$$

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Summary

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